AP011800.1:121836-137709
Candidatus Acetithermum autotrophicum

Ribonuc_L-PSP FUSC Phage_integrase Phage_int_SAM_1 CorA Methyltransf_25 GATase_6 GXGXG CopG_antitoxin BrnT_toxin CutA1 ? Fer4 Glu_synthase Fer4 Glu_synthase FAD_binding_4 FAD-oxidase_C Fer4_8 Pribosyltran Penicil_amidase

AMFJ01002405.1:487-8787
uncultured bacterium

tRNA-synt_2b HGTP_anticodon ? BrnT_toxin CopG_antitoxin DUF34_NIF3 ? YceG_bac GTP_EFTU EF-G_D2 EFG_C BipA_C ? ?

DF820471.1:317714-331224

Candidatus Vecturithrix granuli

BPD_transp_1 NMT1 ? Memo DS ADH_Fe_C Fe-ADH CopG_antitoxin BrnT_toxin DUF4365 ? ? ? ? FGGY_N FGGY_C FGGY_N FGGY_C

BMFB01000001.1:506416-516961
Glycocaulis alkaliphilus

RHH_1 Cpn60_TCP1 Cpn10 Gly-zipper_Omp OmpA Xre-MbcA-ParS_M Xre_MbcA_ParS_C RES EAL BrnT_toxin CopG_antitoxin ? Hydrolase Sod_Fe_C Sod_Fe_N ? Hpt CM_2

BSSF01000019.1:121050-131249
Microbispora sp. NBRC 16548

? MFS_1 GntR FCD Phage_integrase ? ? PIN BrnT_toxin CopG_antitoxin ? AbiJ_NTD4 HTH_31 DUF5753 His-Kinase ? Sigma70_r4

DLVD01000042.1:25144-34722
Peptococcaceae bacterium

GalP_UDP_transf GalP_UDP_tr_C Arm-DNA-bind_4 Phage_int_SAM_3 Phage_integrase HTH_17 Phage_integrase Phage_int_SAM_1 HTH_17 MCPsignal MCPsignal CopG_antitoxin BrnT_toxin Zn_Ribbon_TF TPRs Cu_amine_oxidN1 ResIII Helicase_C

DMMY01000003.1:981-7808
Elusimicrobia bacterium

MazE_antitoxin PemK_toxin BrnT_toxin CopG_antitoxin BrnT_toxin CopG_antitoxin BrnT_toxin CopG_antitoxin PhdYeFM_antitox ParE_toxin NUDIX cNMPBD Patatin

DNSN01000015.1:863-34174
Candidatus Omnitrophota bacterium

? HicA_toxin HicB_lk_antitox PIN CopG_antitoxin BrnT_toxin

DPGQ01000034.1:1193-3613
Actinomycetota bacterium

? VanZ CopG_antitoxin BrnT_toxin ?

DPTL01000095.1:733-6640
Elusimicrobia bacterium

ABC_tran DUF5679 NUDIX His_Phos_1 DUF1304 SBF CopG_antitoxin BrnT_toxin DUF5615 DUF433 Radical_SAM Radical_SAM

DRFC01000159.1:2134-2094
Nitrospirae bacterium

BrnT_toxin CopG_antitoxin ?

DASSSG010000667.1:174-2882
Bryobacteraceae bacterium

CopG_antitoxin BrnT_toxin Cas_Cas1 CRISPR_Cas2 PDDEXK_1

DASTMI010000766.1:1-6563
Longimicrobiaceae bacterium

Amidohydro_3 CopG_antitoxin BrnT_toxin ABC2_membrane_4 Peptidase_M1 ABC_tran

DASTZM010000090.1:6772-19476
Chloroflexota bacterium

MacB_PCD FtsX MacB_PCD FtsX ABC_tran PadR Vir_act_alpha_C HD MR_MLE_N MR_MLE_C Adenyl_transf CopG_antitoxin BrnT_toxin DNA_ligase_A_N DNA_ligase_A_M DNA_ligase_A_C PIN RHH_1 His-Kinase His-Kinase Receiver Receiver Hpt

DASYVT010000057.1:5558-8499
Candidatus Methylomirabilota bacterium

Bac_luciferase BrnT_toxin AMP-binding AMP-binding_C DUF308 DUF308 DUF308 MaoC_dehydratas DUF3750 FmdA_AmdA FmdA_AmdA Radical_SAM Metallophos_2

DASYRT010000210.1:4960-19503
Candidatus Methylomirabilota bacterium

Fer4_7 Oxidored_q3 Oxidored_q2 Proton_antipo_M Proton_antipo_N Proton_antipo_M Proton_antipo_M CopG_antitoxin BrnT_toxin Methyltransf_3 SBP_bac_5 FTCD_N FTCD AIRS_C AIRS G_glu_transpept ? GLUCM-like_C

DASYVD010000149.1:68-5307
Actinomycetota bacterium

? CopG_antitoxin BrnT_toxin ? BNR_2 Uma2 Uma2

DREL01000139.1:1-2436
Phycisphaerales bacterium

Flavokinase BrnT_toxin CopG_antitoxin DHH DHHA1 ?

DATAMC010000138.1:34758-52166
Thermodesulfobacteriota bacterium

? Methyltransf_11 Cupin_2 FAD_binding_4 MS_TIM-barrel MS_C ICL ICL DUF433 DUF5615 BrnT_toxin CopG_antitoxin HepT-like Glycos_transf_1 Glyco_transf_4 AllH ? Pyr_redox_dim Pyr_redox_2 Radical_SAM

DASUFU010000111.1:1-11827
Actinoplanes sp.

GCS2 Glycos_transf_2 ? ? BrnT_toxin ? HKD HKD Aminotran_1_2 DUF4142 DLH

DUEW01000155.1:1-4654
Anaerolineae bacterium

Transposase_20 DEDD_Tnp_IS110 Methyltransf_25 ? Sacchrp_dh_NADP CopG_antitoxin BrnT_toxin

DRNM01000300.1:1-4430
Thermodesulfatator sp.

Methylase_S DUF1016_N YhcG_C HsdM_N N6_Mtase CopG_antitoxin BrnT_toxin

DULK01000079.1:12388-21430
Syntrophomonadaceae bacterium

? NUDIX QueC DrHyd PfkB ? Polysacc_synt ? BrnT_toxin CopG_antitoxin ? ? Germane RHH_1 PIN HicB_lk_antitox

DTXA01000390.1:391-1175
Candidatus Latescibacterota bacterium

BrnT_toxin CopG_antitoxin ?

DTYI01000231.1:3906-3841
Thiotrichales bacterium

BrnT_toxin ResIII UvrB_inter Helicase_C UvrB UVR Aminotran_1_2

DUZJ01000034.1:22582-33537
Planctomycetota bacterium

TMEM260-like Cupin_3 PMI_typeI_cat ? Peptidase_C26 FAD_syn Flavokinase ? DJ-1_PfpI BrnT_toxin CopG_antitoxin DHH DHHA1 ? SBP_bac_10 N_methyl IU_nuc_hydro Glyco_hydro_65C

DQIN01000525.1:212-3190
Candidatus Methanophagaceae archaeon SCGC AAA252-L18

CopG_antitoxin BrnT_toxin Epimerase ? ? ?

DATFNF010000003.1:4943-20914
Trueperaceae bacterium

? ? ? Acetylase S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C Porphobil_deam Porphobil_deamC DsbD_TM BrnT_toxin CopG_antitoxin EFG_C EFG_IV EFG_III EF-G_D2 GTP_EFTU CYTH AMP-binding AMP-binding_C Unstab_antitox adh_short_C2 COX1

DATEGF010000372.1:4549-18151
Stellaceae bacterium

Queuosine_synth RF-1 Glyco_transf_9 G_glu_transpept G_glu_transpept Aminotran_5 Sulfatase PhyH BrnT_toxin CopG_antitoxin Aminotran_1_2 TatA_B_E Beta-lactamase ? PIN ?

DATEFL010000893.1:1-1936
Candidatus Binatia bacterium

Amidohydro_1 HUTI_composite_bact BrnT_toxin CopG_antitoxin

DATEUO010000158.1:397-1945
Candidatus Hodarchaeales archaeon

BrnT_toxin CopG_antitoxin DUF433 DUF5615

DATETQ010000177.1:1-3144
Thermodesulfovibrionales bacterium

TPRs ? BrnT_toxin GmrSD_N ?

DATDRI010000311.1:68-4215
Vicinamibacteria bacterium

HNH_4 HIT MazG-like HNH_2 HTH_PafC BrnT_toxin CopG_antitoxin

DATGDU010000857.1:2322-4416
Frankiaceae bacterium

Succ_DH_flav_C FAD_binding_2 Methyltransf_19 ? RHH_10 BrnT_toxin ?

DAOLBF010000130.1:324-1663
Saprospiraceae bacterium

CopG_antitoxin BrnT_toxin DUF3566

DAOLBS010000047.1:19642-25040
Kiritimatiellia bacterium

DUF1841 ParE_toxin Unstab_antitox ParE_like ? ? CopG_antitoxin BrnT_toxin PIN FitA-like_RHH ? ?

DAOSKU010000060.1:7553-9402
Spirochaetota bacterium

DUF2089 Zn_ribbon_DUF2089 PEPCK_N PEPCK_GTP BrnT_toxin CopG_antitoxin ? ? TIG TIG ? Radical_SAM

DAOOQQ010000001.1:101268-118189
Candidatus Ozemobacteraceae bacterium

Acetate_kinase REase SWI2_SNF2 T1R_D3-like T1RH-like_C Methylase_S Methylase_S ? PemK_toxin ? CopG_antitoxin BrnT_toxin HsdM_N N6_Mtase Zn_ribbon_3 Zn_Ribbon_1 ? tRNA-synt_2 N-OB Zn_ribbon_4 FecR

DATIAQ010000293.1:88-3672
Thermodesulfobacteriota bacterium

TetR_C_7 TetR_N OEP OEP BrnT_toxin CopG_antitoxin H_PPase

DATISF010000059.1:148-8257
Thermoanaerobaculia bacterium

? ? CopG_antitoxin BrnT_toxin Rap1a BrnT_toxin RibD_C Peptidase_S9 Pyridox_ox_2 ? ABC_tran

DATIIO010000075.1:1-7931
Candidatus Saccharimonadales bacterium

? ? ? BrnT_toxin CopG_antitoxin His-Kinase His-Kinase PAS ? ? Receiver ? Arg_tRNA_synt_N tRNA-synt_1d

DATJRI010000154.1:10725-26864
Terracidiphilus sp.

HSP20 ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N Clp_N ? DLH GSHPx ABC_tran ABC_membrane BrnA_antitoxin BrnT_toxin MutS_V MSSS Smr ? ? Lipase_3 RHH_1 HVO_A0114 NTP_transf_2 ParD_antitoxin

DATJNW010000116.1:5630-19867
Bryobacteraceae bacterium

SMC_ScpA SMC_ScpB S4 PseudoU_synth_2 ? DUF192 PMSR GFO_IDH_MocA_C3 GFO_IDH_MocA BrnT_toxin CopG_antitoxin MFS_1 HlyD_3 Biotin_lipoyl_2 OEP OEP STAS HTH_8 AAA_lid_14 NtrC-AAA GAF_2 CarboxypepD_reg OMP_b-brl_4

DATKEJ010000352.1:4835-16249
Rhodothermales bacterium

Uma2 Alpha-amylase SusG_C Por_Secre_tail Glyco_hydro_53 Por_Secre_tail Xre-like-HTH Xre_MbcA_ParS_C CopG_antitoxin BrnT_toxin Cellulase Por_Secre_tail PorV ? ?

DATIZS010000099.1:32-4214
Rhizomicrobium sp.

Cytochrom_C ETF ETF_alpha ETF 3HCDH 3HCDH_N CopG_antitoxin BrnT_toxin Beta-lactamase

DATKOZ010000005.1:82166-99939
Candidatus Binataceae bacterium

PNPOx_N PNP_phzG_C Kelch_KLHDC2_KLHL20_DRC7 Kelch_4 GT87 UB2H Transgly Transpeptidase Adenosine_kin ? CopG_antitoxin BrnT_toxin ? NTP_transferase ManC_GMP_beta-helix PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV adh_short Rce1-like Ig_CFAP65_4th ASH Ig_CFAP65_4th ASH SBBP Lip_A_acyltrans

DATKZY010000214.1:4024-20791
Planctomycetota bacterium

Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N MazE_antitoxin PIN PemK_toxin ? ? ? BrnT_toxin ? AP_endonuc_2 Glyco_hydro_42 HEPN Beta_helix CBM60 Glyco_hydro_10 SGSH_C Sulfatase

DASZUI010000277.1:1-5253
Devosia sp.

QueG_DUF1730 CopG_antitoxin BrnT_toxin GST_N_2 GST_C NAD_binding_10 3Beta_HSD Hydrolase_4 IF3_C IF3_N Ribosomal_L35p Ribosomal_L20

DATMUS010000486.1:404-2719
Candidatus Methylomirabilis sp.

CopG_antitoxin BrnT_toxin ? adh_short_C2 adh_short_C2

DATMNI010000243.1:1675-15441
Candidatus Acidiferrales bacterium

DLH PQQ_2 PQQ_2 PQQ Cytochrome_CBB3 DUF3800 ? CopG_antitoxin BrnT_toxin PQQ_2 DUF3471 ANKs PQQ_2 PQQ_2 YafQ_toxin RelB ? ?

DATNCC010000129.1:18619-31764
Dongiaceae bacterium

ABC_membrane ABC_tran Glycos_transf_2 Glyco_transf_9 Kdo Glyco_transf_9 LpxK CopG_antitoxin BrnT_toxin Hydrolase_4 ? HSP70 HSP70 ? Nitroreductase Nitroreductase PLDc_N DUF4124 ?

DASQAZ010000003.1:1-15851
Bacteroidota bacterium

Glyco_hydro_18 Big fn3 fn3 fn3 Glyco_hydro_8 fn3 Por_Secre_tail CopG_antitoxin BrnT_toxin Por_Secre_tail Peptidase_S74 His-Kinase HisKA_3 Y_Y_Y Reg_prop Reg_prop Reg_prop Reg_prop GerE Receiver

DASPZA010000482.1:205-2571
Pyrinomonadaceae bacterium

Prok-JAB BrnT_toxin CopG_antitoxin tRNA-synt_2

DASRID010000634.1:6763-13210
Chloroflexota bacterium

Sulfatase HipA_2 DUF3037 BrnT_toxin DUF6941 MFS_1 Spermine_synth ? Methyltransf_25 PPDK_N PPDK_N PEP-utilizers MFS_1 MFS_1

DASRUG010000413.1:1-3811
Candidatus Methylomirabilis sp.

ParE_toxin PhdYeFM_antitox SnoaL_2 PhosphMutase CopG_antitoxin BrnT_toxin MoCF_biosynth MoeA_C

QQUC01000011.1:170993-188643
Calditrichaeota bacterium

? zf-FPG_IleRS H2TH Fapy_DNA_glyco ? SpoIIE ECH_1 Peptidase_M16_C Peptidase_M16_C Peptidase_M16 FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP_3 FG-GAP FG-GAP_3 FG-GAP FG-GAP_3 FG-GAP FlgD_ig BrnT_toxin CopG_antitoxin ? Abhydrolase_1 ? Acetylase ? ADH_N AlaDh_PNT_C ADH_zinc_N_2

WOVA01000046.1:2085-16940
Elusimicrobia bacterium

Aconitase dUTPase Radical_SAM ATP-cone NRDD Glrx-like Zn_ribbon_NrdR ATP-cone Aldedh CopG_antitoxin BrnT_toxin ? ? PAS PAS His-Kinase Receiver Trans_reg_C RecG_dom3_C Helicase_C DEAD RecG_wedge ? Receiver

QBUU01000153.1:829-6818
Candidatus Methanogaster sp.

Aconitase Aconitase UPF0175 PIN-TPR-GreABC BrnT_toxin CopG_antitoxin Aconitase ? HEATs HEATs HEATs HEATs SesA TIR ? ? ?

LBXN01000018.1:1-7895

Candidatus Gottesmanbacteria bacterium GW2011_GWC2_39_8

? CopG_antitoxin BrnT_toxin ? MFS_3 TrmB ? EamA EamA adh_short ?

LCQE01000002.1:6351-13519
Microgenomates group bacterium GW2011_GWC1_49_7

? DUF11 Sortase ? Peptidase_M16 Peptidase_M16_C Peptidase_M16_C ? BrnT_toxin CopG_antitoxin Trypsin ? NDK

LJNN01000007.1:108-16180

Anaerolineae bacterium SG8_19

YafQ_toxin RelB DUF234 HTH_5 ATPase_2 ? ? ? CopG_antitoxin BrnT_toxin Sugar-bind HTH_23 AP_endonuc_2 SIS ADH_zinc_N ADH_N SAF DRL_cat NAD_binding_3 FGGY_C FGGY_N Peripla_BP_4

LDXN01000003.1:36490-52107
Candidatus Dadabacteria bacterium CSP1-2

dCMP_cyt_deam_1 RibD_C CTP_transf_like RsfS AhpC-TSA TrmE_N MnmE_helical Aminotran_4 Chorismate_bind BrnT_toxin CopG_antitoxin ? Fer4_ETF_QO ETFQO_UQ-bd PDH_C PDH_N Aminotran_1_2 ACT PDT CM_2 BFN_dom

LGFA01000054.1:241-8409

Moorella sp. 60_41

Peptidase_M41 AAA_lid_3 AAA ? TRAM_2 Radical_SAM UPF0004 RodZ_C HTH_25 FtsK_gamma FtsK_SpoIIIE FtsK_alpha FtsK_4TM YlzJ CLP_protease BrnT_toxin CopG_antitoxin ?

LGFR01000156.1:1-3420

Clostridia bacterium 62_21

TraG-D_C HerA_C CopG_antitoxin BrnT_toxin AAA_25 ? HTH_17 Phage_int_SAM_3

NYYP01000016.1:68139-72796
Opitutia bacterium

DUF2805 ParE-like_toxin ParD_like YafQ_toxin RelB ? ? BrnT_toxin CopG_antitoxin ? ?

PAJQ01000040.1:31164-44351
Chloroflexota bacterium

TENA_THI-4 GYD NMT1 BPD_transp_1 ABC_tran EPSP_synthase HMGL-like BrnT_toxin CopG_antitoxin ? ? Acyl-CoA_dh_2 Acyl-CoA_dh_N Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10 Pribosyltran DZR_2 NUDIX Zn_ribbon_Nudix

NDHX01000013.1:22581-30527
bacterium Unc6

tRNA-synt_2b HGTP_anticodon PPDK_N PPDK_N PPDK_N PEP-utilizers PEP-utilizers_C ? DUF5678 Asp_protease_2 Polysacc_synt Polysacc_synt_3 BrnT_toxin CopG_antitoxin

JACIYF010000083.1:2545-11073
Clostridia bacterium

TrmO_N CbiA Flavin_Reduct Peripla_BP_2 FecCD ABC_tran CopG_antitoxin BrnT_toxin Spore_YunB tRNA-synt_1b S4 ?

JACNKE010000392.1:64-3825
Anaerolineales bacterium

ACAS_N AMP-binding CopG_antitoxin BrnT_toxin Succ_CoA_lig Ligase_CoA_2

JACNHB010001575.1:1-3263
Deltaproteobacteria bacterium

FlgD_ig BrnT_toxin CopG_antitoxin ? HEPN Peptidase_S41

JACNGL010000131.1:310-3782
bacterium

BrnT_toxin CopG_antitoxin Hydrolase ? YjeF_N Carb_kinase

WJLZ01000113.1:359-2636
Chitinivibrionales bacterium

? CopG_antitoxin BrnT_toxin Unstab_antitox Y1_Tnp

JACSSI010000049.1:1-4160
Anaerolineae bacterium

ArsA_ATPase ArsA_ATPase BrnT_toxin CopG_antitoxin Zn_ribbon_8 ? FeS_assembly_P HTH_5 MerR_1 MerR-DNA-bind ?

JADBKR010000006.1:94639-105714
Limnochordaceae bacterium

Acetylase Cys_Met_Meta_PP CMD Phage_integrase Phage_integrase Trans_reg_C CopG_antitoxin BrnT_toxin ? ? Peptidase_M55 CoA_binding_2 Succ_CoA_lig Ligase_CoA_2 ROK ? GFO_IDH_MocA

JADGCS010000014.1:43058-54036
Deltaproteobacteria bacterium

PDZ_6 PDZ_6 Trypsin PDZ_6 Trypsin ? ? ? ? CopG_antitoxin BrnT_toxin Adenine_glyco SMI1_KNR4 ?

JACOSH010000272.1:9277-19351
Acidobacteria bacterium

Sigma70_r2 Sigma70_r4_2 zf-HC2 Peptidase_S8 ? CsgG TPRs TPRs TPRs TPRs TPRs TPRs FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 UnbV_ASPIC BrnT_toxin CopG_antitoxin Sigma70_r2 Sigma70_r4_2 AhpC-TSA DUF362

JACOUC010000005.1:235-1313
Chlamydiae bacterium

CopG_antitoxin BrnT_toxin Phage_integrase

JACOUC010000009.1:52504-58280
Chlamydiae bacterium

His-Kinase His-Kinase MASE8 His-Kinase His-Kinase MerR_1 DUF4258 MqsA_antitoxin NTP_transf_2 HepT-like BrnT_toxin CopG_antitoxin ? DUF2283

JACOUD010000004.1:43054-52395
Chlamydiae bacterium

CopG_antitoxin ? Abhydrolase_1 DUF4410 OMP_b-brl OMP_b-brl ? CopG_antitoxin BrnT_toxin HTH_3 Fic HATPase_c_3 ? DUF3841 PhdYeFM_antitox ParE_toxin

JACOTU010000010.1:1-5392
Candidatus Poribacteria bacterium

His-Kinase HisKA_3 PAS PAS Receiver BPD_transp_1 BrnT_toxin CopG_antitoxin PhyH

JACOTU010000775.1:1337-9514
Candidatus Poribacteria bacterium

CopG_antitoxin BrnT_toxin Rieske ANKs Sigma70_r4_2 Sigma70_r2 ? ? ? ? ANKs Sigma70_r4_2 Sigma70_r2

JACOTU010000788.1:1-4431
Candidatus Poribacteria bacterium

HicA_toxin dnstrm_HI1420 UPF0175 PIN DUF3368 ? PEP-CTERM DUF3800 BrnT_toxin CopG_antitoxin

JACOVX010000003.1:1-3165
Elusimicrobia bacterium

? CopG_antitoxin BrnT_toxin Transket_pyr Transketolase_C Transketolase_N

JACOYE010000007.1:70271-81233
Candidatus Microgenomates bacterium

? ? DUF5667 FtsH_ext AAA AAA_lid_3 Peptidase_M41 tRNA_bind ? CopG_antitoxin BrnT_toxin tRNA-synt_1b CopG_antitoxin BrnT_toxin ? ? Trp_repressor ?

JACPCG010000067.1:2221-3249
Candidatus Rokuibacteriota bacterium

MM_CoA_mutase CoA_transf_3 B12-binding CopG_antitoxin BrnT_toxin

JACPCR010000113.1:8804-25539
Planctomycetota bacterium

tRNA-synt_2e MoCF_biosynth ? MoaC ? Peripla_BP_3 GntR ? BrnT_toxin CopG_antitoxin GmrSD_N GmrSD_C DUF5655 CBM9_1 Xre-like-HTH GCE_fung PhoPQ_related Hydrolase_4 FtsK_4TM AP_endonuc_2

JACPCW010000043.1:14956-12518
Candidatus Rokuibacteriota bacterium

? AI-2E_transport Cation_ATPase_C HAD Cation_ATPase E1-E2_ATPase Cation_ATPase_N ? ? ? DUF3854 BrnT_toxin ? Rad52_Rad22 CBS CBS ? LysR_substrate AIRC NTP_transferase

JACPDT010000058.1:19584-32401
Armatimonadetes bacterium

RadC UPF0758_N TIG TIG NHL SLT ? ? Uma2 CopG_antitoxin BrnT_toxin ? HD_5 ? ? Pribosyltran HTH_WhiA LAGLIDADG_WhiA WhiA_N Gp_dh_C Gp_dh_N

JACPEZ010000020.1:1-11548
Candidatus Daviesbacteria bacterium

HTH_25 Ser_hydrolase ? Sua5_yciO_yrdC ? CopG_antitoxin BrnT_toxin FtsK_alpha FtsK_SpoIIIE FtsK_SpoIIIE FtsK_gamma Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C AAA Lon_lid RNase_PH RNase_PH_C PNPase RNase_PH RNase_PH_C KH_1 S1 Ribosomal_S15 ? ?

JACPEY010000021.1:139-5361
Candidatus Daviesbacteria bacterium

BrnT_toxin CopG_antitoxin ? ? ? Glyco_transf_4 Glycos_transf_1 Glycos_transf_1 FBPase_glpX

JACPFG010000048.1:7975-15630
Deltaproteobacteria bacterium

ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N Clp_N Radical_SAM AzlC AzlD DUF4143 AAA_14 PDZ CopG_antitoxin BrnT_toxin

JACPGD010000307.1:3893-18701

Candidatus Hydrogenedentota bacterium

RecA RecA_C YhdP ? SPOR ApbA ApbA_C DUF374 ? BrnT_toxin CopG_antitoxin zf-B_box adh_short CoA_transf_3 Hpt Receiver Receiver His-Kinase His-Kinase HAMP CHASE8 ? ?

JACPIY010000056.1:2078-13808
Deltaproteobacteria bacterium

AAA_3 VWA_2 Calcineurin Radical_SAM UPF0004 tRNA_Me_trans_C tRNA_Me_trans_M tRNA_Me_trans RHH_3 PIN BrnT_toxin CopG_antitoxin Ion_trans_2 Pps Pps Amino_oxidase

JACPKD010000002.1:379-14014
Candidatus Blackburnbacteria bacterium

? RVT_1 bAvd ? PfkB 2-Hacid_dh_C CopG_antitoxin BrnT_toxin Bro-N Transgly Transpeptidase Transgly Transpeptidase DUF5667 Ribonuc_red_lgC Ribonuc_red_lgN

JACPKD010000032.1:1-8299
Candidatus Blackburnbacteria bacterium

Trypsin Glyco_transf_4 Glycos_transf_1 ? Fapy_DNA_glyco H2TH zf-FPG_IleRS SMC_N ? BrnT_toxin CopG_antitoxin Sigma70_r4 Carb_kinase

JACPOO010000002.1:9772-17954
Chloroflexi bacterium

? ? CofC CofD ? ? PNPOx_N BrnT_toxin Thiolase_C Thiolase_N UPF0175 UPF0175

JACPRC010000054.1:1-9009
Planctomycetota bacterium

Glyco_trans_2_3 Glyco_transf_9 Glyco_transf_4 Glycos_transf_1 ATPgrasp_Ter Methyltransf_11 Glyco_trans_4_4 Glyco_trans_1_4 FGS BrnT_toxin CopG_antitoxin RuvA_N HHH_5 RuvA_C TPRs TPRs

JACPQO010000174.1:1-6149
Chloroflexi bacterium

UvrC_RNaseH_dom HHH_5 BrnT_toxin CopG_antitoxin Hydrolase ROK ROK Cupredoxin_1 DUF72 PCMT Phage_int_SAM_1 Phage_integrase

JACPRP010000071.1:13161-23638
Chloroflexi bacterium

AMP-binding AMP-binding_C_2 ATC_hydrolase Isochorismatase AcetylCoA_hyd_C AcetylCoA_hydro Cytochrom_C Fe-ADH ADH_Fe_C CopG_antitoxin BrnT_toxin ? ? ?

JACPSX010000026.1:2672-7756
Candidatus Tectomicrobia bacterium

PDDEXK_4 HTH_31 ? AbiGi HicA_toxin ? CopG_antitoxin BrnT_toxin Transgly

JACPTR010000297.1:236-8974
Deltaproteobacteria bacterium

BrnT_toxin CMD ? Peptidase_S74 OCD_Mu_crystall ? CoA_transf_3 5_nucleotid_C Calcineurin

JACPTZ010000170.1:4982-4935
Planctomycetota bacterium

Hexapep Hexapep Hexapep ? BrnT_toxin CopG_antitoxin GIY-YIG

JACPUB010000281.1:4839-17351
Candidatus Rokuibacteriota bacterium

Phosphonate-bd GKRP_SIS_N SIS GKRP-like_C ? NamZ_C NamZ_N MFS_1 Glyco_hydro_3 CopG_antitoxin BrnT_toxin AAA_14 DUF4143 Cation_efflux ZT_dimer SSF BcrAD_BadFG ? ? ?

JACPVI010000075.1:67235-17860
Myxococcales bacterium

? Cu-binding_MopE Receiver GAF His-Kinase His-Kinase Receiver RVT_1 ? HEATs BrnT_toxin CopG_antitoxin Amidohydro_2 K319L-like_PKD GH115_C NTP_transf_8 ? Sigma70_r2 Sigma70_r4_2 ? Caspase

JACPXA010000031.1:2390-5895
Elusimicrobia bacterium

? HMA Fer2_BFD zf_CopZ MerR-DNA-bind MerR_1 ? PMT LmeA BrnT_toxin CopG_antitoxin

JACPXA010000133.1:3659-10879
Elusimicrobia bacterium

Acyl_transf_2 Methyltransf_23 HisKA_7TM GAF_3 His-Kinase Receiver LexA_DNA_bind PIN ? ? CopG_antitoxin BrnT_toxin

JACPXI010000987.1:163-5081
Nitrospinota bacterium

? CopG_antitoxin BrnT_toxin AbiJ_NTD4 PG_binding_1 ? Abhydrolase_6 ? Amidohydro_2

JACPYO010000122.1:1-4137
Candidatus Solibacter usitatus

Amino_oxidase ? BrnT_toxin CopG_antitoxin ? Beta-lactamase

JACPYO010000353.1:5500-20765
Candidatus Solibacter usitatus

Calcineurin DUF4432 Patatin Patatin DUF362 AhpC-TSA Sigma70_r2 Sigma70_r4_2 CopG_antitoxin BrnT_toxin FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 UnbV_ASPIC TPRs TPRs TPRs TPRs TPRs TPRs CsgG CsgG ? Peptidase_S8

JACQCG010000166.1:2861-3991
Candidatus Omnitrophota bacterium

CM_2 PIN BrnT_toxin CopG_antitoxin ? BrnT_toxin CopG_antitoxin

JACQGE010000189.1:3347-15470
Acidobacteria bacterium

Glyco_tranf_2_3 ? ? AMP-binding Acyltransferase MFS_1 UbiA Asn_synthase AIRC BrnT_toxin CopG_antitoxin RelB-like ? ? ADSL_C Lyase_1

JACQJW010000006.1:10314-21688
Candidatus Gottesmanbacteria bacterium

MazE_antitoxin PIN tRNA-synt_2 23S_rRNA_IVP SurA_N_3 ? CopG_antitoxin BrnT_toxin ROK HhH-GPD ? Ribosomal_L25p Ribosomal_TL5_C DUF4012 RVT_3 Polysacc_synt

JACQTY010000022.1:6784-10504
Chloroflexi bacterium

His-Kinase HisKA_3 PAS Radical_SAM Fer4_7 Gly_radical PFL-like BrnT_toxin NikA-like SBP_bac_6 SBP_bac_11 Cupin_2 Cupin_2

JACQQW010000103.1:22-7841
Candidatus Omnitrophota bacterium

tRNA-synt_2 N-OB FtsX MacB_PCD CopG_antitoxin BrnT_toxin ABC_tran Aminotran_4 UVR ATP-gua_Ptrans AAA_16 Clp_N Clp_N

JACQPJ010000006.1:21259-30841
Elusimicrobia bacterium

PemK_toxin EamA EamA THF_DHG_CYH THF_DHG_CYH_C FtsJ S4 ? DAO CopG_antitoxin BrnT_toxin ABC_tran ABC_membrane MerR_1 HMA MerT zf_CopZ Methyltransf_25

JACQOY010000029.1:58-14290
Deltaproteobacteria bacterium

PGPGW DUF1841 TRAM Radical_SAM UPF0004 BFN_dom VanZ Hexapep Hexapep CopG_antitoxin BrnT_toxin ? ParE_toxin HTH_20 Radical_SAM Zn_ribbon_HMPTM CDP-OH_P_transf Thioredox_DsbH S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12

JACQUQ010000010.1:427-12274
Chloroflexi bacterium

Thiolase_C_1 Glutaredoxin CoA_transf_3 CopG_antitoxin BrnT_toxin CoA_transf_3 DUF1828 ELFV_dehydrog TrkA_N ELFV_dehydrog_N TatC DDE_3 HTH_33 DDE_5 DDE_5

JACQUS010000117.1:2407-15644
Chloroflexi bacterium

Bac_luciferase CoA_transf_3 BrnT_toxin MmgE_PrpD_N MmgE_PrpD_C Hydantoinase_B Hydant_A_N Hydantoinase_A Hydant_A_C 2-Hacid_dh_C PPC Peptidase_S8 CarboxypepD_reg ? Amidase

JACQVR010000110.1:1-5523
Planctomycetota bacterium

HlyD_D23 ACR_tran CopG_antitoxin BrnT_toxin RHS_repeat RHS_repeat RsmI_C

JACQVX010000192.1:1-13218
Planctomycetota bacterium

ECH_1 3HCDH 3HCDH 3HCDH_N Amidohydro_3 Amidohydro_1 Amidohydro_3 Amidohydro_1 ? FumaraseC_C Lyase_1 CopG_antitoxin BrnT_toxin adh_short Abhydrolase_6 DUF5817

JACQVW010000052.1:137-6776
Planctomycetota bacterium

DUF11 NPCBM CopG_antitoxin BrnT_toxin Biotin_carb_N CPSase_L_D2 Biotin_carb_C HMGL-like PYC_OADA Biotin_lipoyl Zn_ribbon_TFIIB

JACQVW010000762.1:17262-11672
Planctomycetota bacterium

? DUF4058 AAA_14 DUF4143 adh_short_C2 BrnT_toxin CopG_antitoxin ? PK_C PK Aminotran_1_2 AMP-binding ? Uma2

JACQWW010000120.1:1-10416
Elusimicrobia bacterium

DUF2130 ? ? MtN3_slv DUF2204 ? CopG_antitoxin BrnT_toxin AAA_3 VWA_CoxE ? ? ? AAA_14 DUF4143 ?

JACQXF010000023.1:6001-13375
Nitrospirae bacterium

DUF6765 NAD_Gly3P_dh_C NAD_Gly3P_dh_N HI0933_like Maf Dus DUF2442 CopG_antitoxin BrnT_toxin AF2212-like PIN

JACQXH010000182.1:1-4788
Nitrospirae bacterium

DUF2064 ? SAICAR_synt CopG_antitoxin BrnT_toxin PAS PAS SBP_bac_3

JACQYJ010000042.1:185-6584
Deltaproteobacteria bacterium

CopG_antitoxin BrnT_toxin PDDEXK_1 LpxC ? PhdYeFM_antitox PIN BrnT_toxin ?

JACQYK010000136.1:451-8686
Deltaproteobacteria bacterium

AFOR_C AFOR_N PAP2 CopG_antitoxin BrnT_toxin Zn_ribbon_Thio2 Thioredoxin EamA EamA TatD_DNase Ribonuc_red_lgN Ribonuc_red_lgC TSCPD

JACQYF010000012.1:358-14041
Chloroflexi bacterium

ATP-synt_C ATP-synt_B OSCP ATP-synt_ab_N ATP-synt_ab ATP-synt_ab_C ? ? ATP-synt BrnT_toxin CopG_antitoxin ATP-synt_DE_N ATP-synt_DE MreB_Mbl CoA_binding Put_DNA-bind_N dCache_1 HAMP HisKA_3 His-Kinase Receiver GerE Fer4_8 CCG

JACQYT010000045.1:21335-17510
Elusimicrobia bacterium

? PK ? ? ParA-Soj-NTPase CopG_antitoxin BrnT_toxin NOV_C Helicase_C SNF2-rel_dom His-Kinase STAS_2 DUF4325 PIN ABC-ATPase AAA_23 ? MethyltransfD12 DUF1156

JACQZT010000330.1:1-14698
Acidobacteria bacterium

ACCA PHP DNA_pol3_alpha DNA_pol3_finger HHH_6 N-OB HicB_lk_antitox CopG_antitoxin BrnT_toxin Hydrolase_4 Hydrolase_4 Pur_ac_phosph_N Calcineurin CarboxypepD_reg Plug OMP_b-brl_4 GatB_N GatB_Yqey ? Sulfatase SGSH_C PTA_PTB PTA_PTB

JACRON010000011.1:1-6033
Nitrospirae bacterium

THF_DHG_CYH THF_DHG_CYH_C PsiE GTP_cyclohydroI 5-FTHF_cyc-lig CbiA DUF4143 AAA_14 CopG_antitoxin BrnT_toxin TPRs

JACRJC010000034.1:56547-69227
Rhodospirillales bacterium

Receiver ? TauE Usp Usp Abhydrolase_6 MqsA_antitoxin CopG_antitoxin BrnT_toxin Aminotran_3 AAA_26 Aminotran_1_2 Rhodanese IF3_N IF3_C TGS tRNA_SAD tRNA-synt_2b HGTP_anticodon ?

JACRGV010000103.1:1-5594
Nitrospirae bacterium

HTH_11 Peptidase_M50 DUF2283 AAA CopG_antitoxin BrnT_toxin TonB_C WCX DUF7718 DUF5647 DUF2683 MazE_antitoxin ParE_like ?

JACRKQ010000053.1:59858-73414
Candidatus Saganbacteria bacterium

Fic PhdYeFM_antitox ParE_toxin 2-Hacid_dh_C PGDH_inter ACT Patatin H_PPase tRNA-synt_1 Anticodon_1 zf-FPG_IleRS BrnT_toxin CopG_antitoxin AdoHcyase AdoHcyase_NAD DUF5665 Peptidase_A8 LGT QueC DUF502

JACRCS010000577.1:766-5970
Deltaproteobacteria bacterium

? SLT BrnT_toxin CopG_antitoxin NTP_transf_2 HepT-like_2 Polbeta PIN FitA-like_RHH HTH_Tnp_ISL3 HicB_lk_antitox

JACRPW010000030.1:1655-10312
Deltaproteobacteria bacterium

LPD3 ? CopG_antitoxin BrnT_toxin EPSP_synthase EPSP_synthase Cytidylate_kin LYTB UbiA SCO1-SenC RNaseH_like ABC1

JAEIOF010000155.1:1-4033
Verrucomicrobia bacterium

? BrnT_toxin CopG_antitoxin Amidohydro_1 LGT NAD_kinase_C NAD_kinase

JADJOL010000003.1:680682-698626
Candidatus Leptovillus affinis

UPF0175 ? EAD7 EAD8 Trypsin ? FGS nSTAND1 EAD8 WYL Helicase_C_3 DDE_Tnp_1 BrnT_toxin CopG_antitoxin ? ? Cas10-Cmr2_palm2 Cas10-Cmr2_palm2 ? RAMPs

JADJZR010000021.1:25237-11474
Elusimicrobia bacterium

bAvd RVT_1 FGS BrnT_toxin CopG_antitoxin OmpA DDE_Tnp_1 CopG_antitoxin OmpA ? TraF_2

JADKJT010000037.1:1691-16691
Gemmatimonadaceae bacterium

Cytochrom_C Cu-oxidase_3 ? Unstab_antitox ParE_toxin DDE_Tnp_1_6 DUF772 PhdYeFM_antitox ParE_toxin BrnT_toxin CopG_antitoxin HicA_toxin HicB_lk_antitox CarboxypepD_reg DUF5128 BrnA_antitoxin CarboxypepD_reg CarboxypepD_reg

PBKC01000072.1:18642-34478
Rhodospirillaceae bacterium

MR_MLE_N MR_MLE_C Peptidase_S9_N Peptidase_S9 ? Astacin DUF924 Reprolysin_5 FeeM cNMPBD BrnT_toxin CopG_antitoxin ILV_EDD_C ILVD_EDD_N adh_short ? TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N DUF192 WD40_Prp19 WD40_Prp19

JADHWG010000064.1:214-8866
Candidatus Omnitrophota bacterium

Nitroreductase CopG_antitoxin BrnT_toxin ResIII UvrB_inter Helicase_C UvrB UVR QRPTase_N QRPTase_C S4 PseudoU_synth_2 tRNA-synt_1 Anticodon_1 BRI3BP Val_tRNA-synt_C

JADHXZ010000034.1:1-11183
Phycisphaerae bacterium

FAD_binding_4 Mur_ligase Mur_ligase_M Mur_ligase_C CopG_antitoxin BrnT_toxin Glyco_transf_28 Glyco_tran_28_C FTSW_RODA_SPOVE MraY_sig1 Glycos_transf_4 Mur_ligase Mur_ligase_M Mur_ligase_C Mur_ligase Mur_ligase_M Mur_ligase_C PBP_dimer Transpeptidase ?

VGOB01000020.1:13981-13255
Chloroflexi bacterium

Eco57I MazG Pyr_redox_2 Fer4_20 CBS CBS ELFV_dehydrog ELFV_dehydrog_N ? CopG_antitoxin BrnT_toxin Esterase Esterase Esterase Abhydrolase_1 ? AMMECR1 DUF2442

VGLC01000145.1:4671-18605
Candidatus Poribacteria bacterium

GFO_IDH_MocA GFO_IDH_MocA_C3 Uma2 Hepar_II_III_N Hepar_II_III_C ? Transposase_31 LRRs LRRs LRRs Cohesin FlgD_ig BrnT_toxin CopG_antitoxin ? DUF4340 DUF4340 LamininG LamininG Aldo_ket_red GFO_IDH_MocA_C3 GFO_IDH_MocA

VGEN01000099.1:160-10388
Alphaproteobacteria bacterium

UPF0093 Ferrochelatase URO-D Kinase-PPPase Maf BrnT_toxin CopG_antitoxin DUF1272 CreA Acetylase Acetylase Patatin DUF3376 CoaE

VGYL01001289.1:70-1973
Planctomycetota bacterium

CopG_antitoxin BrnT_toxin ATP-synt_ab ATP-synt_VA_C

VGWY01000203.1:443-4270
Nitrospira sp.

? ? ? CopG_antitoxin BrnT_toxin ACT ACT

VGSF01000172.1:916-4793
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin DUF3485 ? HicB_lk_antitox HicA_toxin ? DUF433

VGOD01000090.1:11550-21302
Chloroflexi bacterium

? ? ParA-Soj-NTPase Ubie_methyltran BPD_transp_1 ATP-synt_DE_N ATP-synt_DE CopG_antitoxin ? ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C ATP-synt ATP-synt_ab_N ATP-synt_ab ATP-synt_ab_C ParE_toxin ?

VGOG01000034.1:15249-23524
Chloroflexi bacterium

FAD_binding_2 Fer2 Fer4_10 CBS CBS TTHA0829-like_ACT FAD_oxidored Fer4_10 ? NAD_binding_8 Fer4 Fer4_6 CopG_antitoxin BrnT_toxin CCG

JAFGEX010000126.1:174-6342
bacterium

PDZ_2 PDZ_2 Sod_Fe_C Sod_Fe_N Amidohydro_1 FTCD_C FTCD FTCD_N PhdYeFM_antitox BrnT_toxin CopG_antitoxin Urocanase_C

JAFGRW010000115.1:122482-139486
Spirochaetes bacterium

? MCPsignal HAMP ? Pirin_C Pirin Acetylase ? H_PPase BrnT_toxin CopG_antitoxin Glug Glug fn3 K319L-like_PKD GSHPx Fic MazE_antitoxin ? TetR_N

JAFKJY010000070.1:10654-21464
Hyphomicrobiales bacterium

DUF1638 DUF4893 ExoD Virulence_fact DUF5981 MTHFR Pterin_bind BrnT_toxin CopG_antitoxin ? Fer2 RACo_linker Raco_middle RACo_C_ter SnoaL_3 Abhydrolase_3 LysR_substrate HTH_1 DUF1127

JABIKI010000037.1:1-7255
Candidatus Scalindua sp.

Methyltransf_28 3keto-disac_hyd SBP_bac_8 BrnT_toxin CopG_antitoxin ? ? Rhomboid BPD_transp_1 BPD_transp_1 ABC_tran ?

JAHJDR010000155.1:285-12063
bacterium

3HCDH_N 3HCDH ECH_1 Thiolase_N Thiolase_C CBS CNNM AhpC-TSA CopG_antitoxin BrnT_toxin DUF6290 BrnT_toxin RHH_1 ? NT5C MgtC Cu-binding_MopE ? ?

JAHIZL010000133.1:7-9423
Candidatus Omnitrophota bacterium

? ? ? CopG_antitoxin BrnT_toxin HicA_toxin HicB_lk_antitox HEAT_EZ Pribosyltran Pribosyltran_N

JAHIRR010000331.1:1-2659
Candidatus Omnitrophota bacterium

CutA1 4HBT DUF481 OsmC ? ? BrnT_toxin

JAHJMP010000288.1:1-6063
bacterium

? Cytochrome_C7 Cytochrome_C554 Shikimate_DH GlutR_N GlutR_N ABC_tran BrnT_toxin CopG_antitoxin Rhodanese Glyco_trans_1_4 Nitrate_red_del ? PIN

JAHJUB010000153.1:1489-13554
Candidatus Omnitrophota bacterium

YajC Ni_insertion Peptidase_C1 S4 PseudoU_synth_2 RrnaAD Enolase_N Enolase_C CopG_antitoxin BrnT_toxin Calcineurin Phage_int_SAM_1 Phage_integrase Lactamase_B Nlig-Ia DNA_ligase_aden DNA_ligase_OB DNA_ligase_ZBD HHH_2 BRCT Cupin_2 Transposase_29

JAHJSU010000104.1:1-5731
Elusimicrobia bacterium

Methyltransf_25 Peptidase_M20 Lysine_decarbox ? BrnT_toxin CopG_antitoxin

JAHJSK010000096.1:360-5498

Patescibacteria group bacterium

CopG_antitoxin BrnT_toxin N_methyl T2SSF T2SSF T2SSE Receiver T2SSE

JAGWZJ010000021.1:18040-23797
Proteobacteria bacterium

Ribosomal_L27 Ribosomal_L21p polyprenyl_synt FlgM ? LIP CopG_antitoxin BrnT_toxin

JAHRVQ010000015.1:392-10777
Pseudomonadales bacterium

HicB UPF0150 HicA_toxin HTH_3 HigB-like_toxin ? BrnT_toxin CopG_antitoxin Polyketide_cyc2 AMP-binding_C AMP-binding adh_short_C2 CoA_transf_3 4HBT_3 ECH_1

JAFAMZ010000239.1:107-8576
Silvibacterium sp.

Rieske TIM BrnT_toxin CopG_antitoxin PGK Ribonuc_L-PSP Gp_dh_C Gp_dh_N DUF3300 DUF2950 Hydrolase_4

JAFATJ010001234.1:76-3075
Acidobacteriaceae bacterium

BrnT_toxin ? Bac_DNA_binding TIR PIN

JAFASG010001146.1:133-2890
Acidobacteriaceae bacterium

BrnT_toxin ? Methylase_S Methylase_S N6_Mtase

JAFDFL010000536.1:333-2972
Deltaproteobacteria bacterium

CopG_antitoxin BrnT_toxin Fer4_10 NrfD

JAFDCM010000277.1:645-4094
Deltaproteobacteria bacterium

Phage_integrase Phage_int_SAM_4 CopG_antitoxin BrnT_toxin PhdYeFM_antitox YoeB_toxin CENP-H

JAFDAD010000139.1:674-4477
Deltaproteobacteria bacterium

BrnT_toxin ? Lon_C DUF3378 RNase_HII

JACFMS010000010.1:24589-36579
Sphingomonadales bacterium

OmpW DUF2207_C DUF2207 LemA Lactamase_B BLACT_WH ? NadA QRPTase_N QRPTase_C BrnT_toxin CopG_antitoxin AsnC_trans_reg HTH_24 Cys_Met_Meta_PP adh_short_C2 ZT_dimer Cation_efflux PNP_phzG_C PNPOx_N Gly-zipper_Omp 17kDa_Anti_2

JAHBVH010000004.1:161490-178759
Caldilineaceae bacterium

ECF-ribofla_trS ECF-ribofla_trS HAMP GNAT_acetyltran TLC AdoMet_dc Peptidase_S8 Peptidase_M20 CopG_antitoxin BrnT_toxin T3SS_needle_E WCX WYL ? Csc2 ? ResIII

JAIVIU010000688.1:1-3007
Thermomicrobia bacterium

? Glyoxalase_6 Glyoxalase MR_MLE_N MR_MLE_C CopG_antitoxin BrnT_toxin

JAGQLU010000089.1:248-2805
Candidatus Woesebacteria bacterium

? MreB-like_C CopG_antitoxin BrnT_toxin

JAGQFX010000096.1:7865-28787
Anaerolineales bacterium

? Pyr_redox_2 Fer4_9 ? Cytochrome_C7 Cytochrome_C7 Mtrc-MtrF_II-IV_dom DUF2937 G_glu_transpept BrnT_toxin CopG_antitoxin DUF433 TPRs TPRs DUF4810 AP-ATPase TPRs TPRs TPRs AP-ATPase N_BRCA1_IG PPC SLH SLH SLH

JAGQFW010000863.1:1-7255
Anaerolineales bacterium

LysM DegT_DnrJ_EryC1 Glycos_transf_2 PfkB AdoHcyase_NAD AdoHcyase PAP2 CopG_antitoxin BrnT_toxin ?

JAGRCT010000436.1:1-2170
Caldilineaceae bacterium

RNA_ligase CopG_antitoxin BrnT_toxin Beta-lactamase

JAGRBE010000389.1:1759-5012
Calditrichaeota bacterium

PFK Sirtuin Sirtuin BrnT_toxin CopG_antitoxin TPRs TPRs TPRs TPRs TPRs

JACKBC010000004.1:71574-89536
Caldilineaceae bacterium

Acetylase BPD_transp_1 SBP_bac_8 GFO_IDH_MocA GFO_IDH_MocA_C3 Methyltransf_25 Dala_Dala_lig_C ? CopG_antitoxin BrnT_toxin SMC_ScpA N_BRCA1_IG Amidase_2 Amidase_2 Amidase_2 ? ECH_1 Flavoprotein AP_endonuc_2 SEC-C YchJ_M-like SEC-C

JAGGSH010000035.1:4692-17176
bacterium

? Glyco_hydro_42 Peptidase_C39_2 EamA ? PCRF RF-1 TPP_enzyme_C CopG_antitoxin BrnT_toxin POR_N PFOR_II ABC2_membrane Ribosomal_L33 TPRs PSP1 EF_TS Ribosomal_S2

JAJSLJ010000068.1:5998-11587
Candidatus Methylacidiphilales bacterium

Nlig-Ia DNA_ligase_aden DNA_ligase_OB DNA_ligase_ZBD HHH_5 HHH_2 BRCT ? CopG_antitoxin BrnT_toxin tRNA_bind B3_4 B5 tRNA_synthFbeta FDX-ACB ? Couple_hipA HipA_C HTH_3 ? CBS Voltage_CLC SpoIIAA-like

JAIPIQ010000003.1:24230-43532
Candidatus Neomarinimicrobiota bacterium

Peptidase_M28 M20_dimer HKD SNF2-rel_dom Helicase_C DUF4062 HATPase_c_4 Fic-like_C ORF6N Peptidase_S26 SnoaL_4 cobW BrnT_toxin CopG_antitoxin bMG10 A2M A2M_BRD bMG6 bMG5 MG2 bMG3

JAIPHQ010000041.1:1-6641
Candidatus Omnitrophota bacterium

AAA_14 DUF4143 AbiEi_3_N AbiEi_3 AbiEii DUF655 BrnT_toxin CopG_antitoxin AbiEi_4 AbiEii tRNA_anti_2 Exonuc_VII_L

JAKLQR010000244.1:7764-17444
Thermodesulfovibrionales bacterium

2Fe-2S_thioredx Fer4_7 NADH_4Fe-4S SLBB Complex1_51K 2Fe-2S_thioredx Molydop_binding cNMPBD FTHFS Hist_deacetyl CopG_antitoxin BrnT_toxin

JADFQP010000283.1:1-2735
Proteobacteria bacterium

AsmA HAD_2 BrnT_toxin CopG_antitoxin Acetyltransf_CG

JADFPH010000116.1:39-7996
Chloroflexota bacterium

MR_MLE_N MR_MLE_C APH MFS_1 MR_MLE_N MR_MLE_C BrnT_toxin CopG_antitoxin Glyoxalase 2-Hacid_dh_C ?

JADFOY010000109.1:8645-6461
Candidatus Neomarinimicrobiota bacterium

ParE_toxin PhdYeFM_antitox HycI ParE_toxin PhdYeFM_antitox CopG_antitoxin BrnT_toxin PemK_toxin MazE_antitoxin ? Rce1-like HTH_31 HigB-like_toxin Gp49

JADFGV010000067.1:4189-13907
Chloroflexota bacterium

? Pribosyltran DZR_2 Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10 Acyl-CoA_dh_2 Acyl-CoA_dh_N DinB_2 ? CopG_antitoxin BrnT_toxin HMGL-like ThiC_Rad_SAM ThiC-associated DUF3800 ?

JALAHS010000060.1:2452-17966
Acidobacteria bacterium

? ? Cas_GSU0054 Cas_GSU0053 ? Cas3_HD Cas3-like_C_2 ResIII WCX WYL BrnT_toxin CopG_antitoxin RHH_1 ? TPRs TPRs AhpC-TSA ? ?

JAKEFQ010000144.1:1-3411
bacterium

SBBP SBBP ? CopG_antitoxin BrnT_toxin CopG_antitoxin BrnT_toxin

JAKEEU010000274.1:13890-26837
candidate division KSB1 bacterium

RES Zincin_1 CT398_CC Zn_ribbon_9 DHH DHHA1 RecJ_OB Carboxyl_trans PHD_like HicA_toxin BrnT_toxin CopG_antitoxin PhdYeFM_antitox ? AtuA NTP_transf_3 Hexapep LbH_EIF2B AtuA_ferredoxin ECH_1

JAJHSH010000066.1:53998-5010
Candidatus Acetothermia bacterium

BrnT_toxin CopG_antitoxin Methyltransf_11 ZT_dimer Cation_efflux ? DOMON DOMON ?

QYKX01000629.1:70-1767
Candidatus Electrothrix sp. EH2

? PIN PIN BrnT_toxin CopG_antitoxin DUF4258

JALHVQ010000450.1:380-2725
Actinomycetota bacterium

? ? ParE_toxin ? BrnT_toxin CopG_antitoxin

JALHWA010000238.1:1-5992
Anaerolineales bacterium

Met_synt_B12 RACo_C_ter Raco_middle RACo_linker Fer2 Phage_int_SAM_3 Phage_integrase CopG_antitoxin BrnT_toxin NAD_synthase NAD_synthase

JAGMAX010000395.1:224-1952
Candidatus Omnitrophota bacterium

? ? DUF1016_N CopG_antitoxin BrnT_toxin

JAGLKT010001197.1:235-880
bacterium

BrnT_toxin CopG_antitoxin

JAKLJX010000012.1:171042-180860
bacterium

? AtuA DUF433 ? DUF6847 ? CopG_antitoxin BrnT_toxin Carboxyl_trans Asp_protease_2 ? ParE_toxin ? DHH DHHA1 RecJ_OB Dala_Dala_lig_C Dala_Dala_lig_N

JAGLCY010000025.1:45-4684
Candidatus Krumholzibacteriia bacterium

? YoeB_toxin HTH_3 CopG_antitoxin BrnT_toxin Phage_int_SAM_4 Phage_integrase HIPIP ?

JAMCTM010000039.1:8-14228
Patescibacteria group bacterium

? Glyco_transf_4 Glycos_transf_1 NYN Glyco_transf_4 ? Methyltransf_11 CopG_antitoxin BrnT_toxin LamininG Glycos_transf_2 ? HicA_toxin ? ? ?

JAMCTB010000110.1:1356-2381
Chloroflexota bacterium

? CbiM BrnT_toxin CopG_antitoxin

JAMCQT010000239.1:199-7740
Patescibacteria group bacterium

Peptidase_M50 RRF NUDIX CopG_antitoxin BrnT_toxin ? EF-Ts_N EF_TS Ribosomal_S2 ? 23S_rRNA_IVP ? AbiEii AbiEii

JAMCPO010000019.1:1-10822
Nitrospirae bacterium

MotA_ExbB CarboxypepD_reg CarboxypepD_reg Plug TonB_dep_Rec_b-barrel CopG_antitoxin BrnT_toxin Transposase_20 Phosphoesterase CarboxypepD_reg CarboxypepD_reg Plug TonB_dep_Rec_b-barrel

JAHDWH010000004.1:44967-64801
Caldilineaceae bacterium

UvrD_C UvrD-helicase Glycos_transf_1 Glyco_transf_4 PIN ? WD40_RLD RCC1_2 WD40_RLD WD40_RLD ParE_toxin CopG_antitoxin BrnT_toxin TPRs TPRs Wzy_C Glycos_transf_4 ? UbiA ? ECH_1 ABC_membrane ABC_tran

JAMCVL010000092.1:1-2736
Nitrospirae bacterium

? ? N_methyl BrnT_toxin CopG_antitoxin

JAMJQR010000010.1:141-7860
Candidatus Benthicola marisminoris

? RtcB BrnT_toxin CopG_antitoxin ? Cys_Met_Meta_PP Peptidase_M14 CarboxypepD_reg DsbD_2 ?

JAMLHS010000026.1:5903-8677
Chloroflexota bacterium

? ? BrnT_toxin CopG_antitoxin ? RuvC ? DUF5652 Pantoate_ligase Pantoate_transf

JAAEOP010000072.1:40-2141
Chloroflexota bacterium

DDE_Tnp_Tn3 BrnT_toxin CopG_antitoxin Resolvase HTH_7

JAAEOW010000620.1:1-9684
Chloroflexota bacterium

DUF937 STY-kinase FGS ? CopG_antitoxin BrnT_toxin Na_H_Exchanger TrkA_N TrkA_C Pyrophosphatase STY-kinase TIR

JANLHZ010000033.1:10242-16216
Planctomycetota bacterium

Peptidase_MA_2 GHMP_kinases_C GHMP_kinases_N MOSC ? ? Glyoxalase CopG_antitoxin BrnT_toxin

JAAXKT010000013.1:72015-84465
Ignavibacteria bacterium

? His-Kinase His-Kinase PAS PAS_10 WEMBL CheR CheR_N CheB_methylest Peptidase_S41 Peptidase_S41 YgaP-like_TM Receiver PAS cNMPBD HTH_Crp_2 BrnT_toxin CopG_antitoxin YfhO

JAGIXR010000231.1:82-4077
Candidatus Electrothrix japonica

? ? BrnT_toxin CopG_antitoxin ? ? Ribosomal_S2 EF-Ts_N EF_TS

JAPLLY010000014.1:306-5627
Candidatus Omnitrophota bacterium

BrnT_toxin CopG_antitoxin DUF6932 HTH_37 HTH_24 DAO NTP_transferase Epimerase

JAPLLS010000040.1:14506-14452
Candidatus Omnitrophota bacterium

SWI2_SNF2 T1R_D3-like T1RH-like_C ? ? GmrSD_N GmrSD_C ? DUF7111 Lipase_3 ORF6N BrnT_toxin CopG_antitoxin Gp49 HTH_3 HKD ORF6N PAS PAS His-Kinase His-Kinase DUF6941

JAPLJI010000128.1:10542-18417
Proteobacteria bacterium

Glyco_hydro_5_C Cellulase DPM2 Fer4_21 Lactamase_B MshEN T2SSE T2SSE cNMPBD BrnT_toxin CopG_antitoxin

JAPLHB010000137.1:22416-14200
Chloroflexota bacterium

Vgb_lyase Vgb_lyase Vgb_lyase PA14 ? Methyltransf_25 LGT AhpC-TSA PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C ? BrnT_toxin AbiEii AbiEii DUF6922 Memo Rhomboid TPRs TPRs Guanylate_kin Zn_ribbon_8 CdAMP_rec

JAPKYU010000403.1:25317-7101
Acidobacteria bacterium

BrnT_toxin CopG_antitoxin GFO_IDH_MocA GFO_IDH_MocA_C2 AP_endonuc_2 dTDP_sugar_isom ? ? GcpE

JAPOOL010000389.1:138-8392
Chloroflexota bacterium

? TauE BrnT_toxin CopG_antitoxin TGT ThiS NIL PALP NinF ? adh_short_C2

JAOSKG010000001.1:1966752-1976434
Melioribacteraceae bacterium

TauE Sua5_yciO_yrdC Sua5_C Inositol_P ? ABC_tran ChlI AAA_25 Zn_ribbon_LapB CopG_antitoxin BrnT_toxin Tll0287-like AhpC-TSA AhpC-TSA DUF1858 DUF6132 ? DPBB_1

JAPZRE010000009.1:43369-58156
Beijerinckiaceae bacterium

Apolipoprotein Apolipoprotein Apolipoprotein Apolipoprotein Apolipoprotein Apolipoprotein Apolipoprotein Hpt Fer2 MFS_1 Usp ? CopG_antitoxin BrnT_toxin VWA_CoxE Bac_DNA_binding FAD_binding_3 FAD_binding_3 PilZ PilZ PilZ Cupin_2

JAPTXQ010000324.1:429-2628
Actinomycetota bacterium

2-Hacid_dh_C BrnT_toxin CopG_antitoxin

JAPTWG010000069.1:3183-24612
Nitrospiraceae bacterium

N_methyl N_methyl GspH ? ? His-Kinase His-Kinase HAMP Receiver HTH_8 AAA_lid_14 NtrC-AAA Receiver BrnT_toxin Alginate_exp ACR_tran ACR_tran HlyD_D23 OEP OEP YtkA Trans_reg_C Receiver His-Kinase His-Kinase HAMP

JAQGKG010000241.1:404-8012
Devosia sp.

IspA SRP54_N SRP54 UPF0004 Radical_SAM DAP_epimerase DAP_epimerase ? Acetyltransf_6 CopG_antitoxin BrnT_toxin SRP54_N

JAQTZV010000006.1:216826-12726
Desulfuromonadaceae bacterium

Methylase_S Methylase_S ? ? FRG BrnT_toxin CopG_antitoxin HVO_A0114 SLFN_AlbA_2 HATPase_c_4 HTH_24 DUF4145 REase ResIII Helicase_C EcoEI_R_C AAA_14 DUF4143 TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C ?

JAQVHF010000037.1:1251-12667
Desulfobulbaceae bacterium

FAD_binding_4 BBE Flavin_Reduct CitMHS CitMHS RHH_1 PemK_toxin DUF4143 AAA_14 DUF2196 BrnT_toxin CopG_antitoxin dnstrm_HI1420 DUF4143 AAA_14 DUF2196 PhdYeFM_antitox ParE_toxin dnstrm_HI1420

JAQTOU010000015.1:675-5348
Elusimicrobiales bacterium

BrnT_toxin CopG_antitoxin ? HD_5 Rrf2 Receiver Receiver Phage_integrase

JAPPKT010000656.1:2621-9187
Caldilineaceae bacterium

UxuA DUF2283 Hydrolase Zn_ribbon_13 ? 7TMR-DISM_7TM Zn_ribbon_3 CopG_antitoxin BrnT_toxin ? adh_short_C2 AP_endonuc_2

JAGWKM010000282.1:273-1334
Patescibacteria group bacterium

DUF6883 DUF4926 BrnT_toxin

JAPPKM010000069.1:4756-20402
Chloroflexota bacterium

ABC_tran ABC2_membrane ABC2_membrane MMPL MMPL MPAB_Lcp_cat Acetylase G_glu_transpept BrnT_toxin CopG_antitoxin Methyltransf_11 ABC2_membrane_3 DrrA1-3_C ABC_tran NHL NHL ? Epimerase

JAGWNE010000846.1:1-1643
Chloroflexota bacterium

? Epimerase BrnT_toxin

JAOUDW010000024.1:1-6456
Nitrospira sp.

BrnT_toxin CopG_antitoxin DUF1810 RelA_SpoT ? DUF2283 ? HEATs HEATs

JARYMM010000017.1:29857-45314
Anaerolineae bacterium

Cytochrom_C_asm AhpC-TSA Cytochrom_NNT Val_tRNA-synt_C Anticodon_1 tRNA-synt_1 PIN HicB tRNA-synt_2c tRNA_SAD DHHA1 BrnT_toxin CopG_antitoxin DHHA1 HicB_lk_antitox Baseplate_J RuvX YceG Peripla_BP_6 Peripla_BP_6

JARYLU010000104.1:133-716
Bacillota bacterium

BrnT_toxin CopG_antitoxin

JASCXX010000001.1:335562-6912
Anaerobaca lacustris

BrnT_toxin CopG_antitoxin GFO_IDH_MocA_C3 GFO_IDH_MocA DHDPS PdxJ T2SSF T2SSF CDP-OH_P_transf DHBP_synthase GTP_cyclohydro2

JASEJK010000131.1:1-3002
Candidatus Omnitrophota bacterium

PGM_PMM_III PGM_PMM_II PGM_PMM_I BrnT_toxin CopG_antitoxin

JASEHD010000188.1:40-3580
Bacteroidota bacterium

TrkH Lyase_aromatic CopG_antitoxin BrnT_toxin

JASEHI010000008.1:30438-9973
Anaerolineales bacterium

Epimerase HTH_24 PMT_2 GDP_Man_Dehyd BrnT_toxin CopG_antitoxin NTP_transferase GDP_Man_Dehyd DegT_DnrJ_EryC1 Acetylase Glycos_transf_2 ?

JASEGY010000074.1:2896-10929
bacterium

CCDC90-like FTHFS ParE_toxin RelB-like Tad CODH_A_N CdhC ACS_CODH_B_C CopG_antitoxin BrnT_toxin Prismane

JASEGT010000034.1:2192-12879
Actinomycetota bacterium

? ? HNH IstB_IS21 HTH_28 ? CopG_antitoxin BrnT_toxin DUF2442 HTH_3 ? HTH_Crp_2 Fic Fic_N Citrate_synt baeRF_family3 ?

JASEIE010000101.1:9103-11057
Thermodesulfobacteriota bacterium

DruA DUF3617 OmpA HEPN NTP_transf_2 CopG_antitoxin BrnT_toxin B12-binding Radical_SAM ? TauE MFS_1 DUF1232 HTH_3 Glycos_transf_2

JASGWF010000049.1:292-13749
Spirochaetota bacterium

MUG113 PIN PIN HicB PhdYeFM_antitox PIN HTH_3 BrnT_toxin CopG_antitoxin Ala_racemase_C Ala_racemase_N GTP_EFTU EF-G_D2 EFG_III EFG_IV EFG_C TonB_2 Virul_fac_BrkB PSII_BNR PSII_BNR PSII_BNR Virul_fac_BrkB PAP2 ROK

SZPG01000016.1:35294-44443
Chloroflexi bacterium CFX6

IMS_C IMS_HHH_1 IMS WCX WYL HTH_11 ? GyrI-like UPF0047 Peptidase_C92 CopG_antitoxin BrnT_toxin DUF2442 NAD_binding_10 Glyco_trans_4_4 Glyco_trans_1_4 OB_ChsH2_C zf-ChsH2

JAUSFA010000095.1:66-3497
Candidatus Daviesbacteria bacterium

? ? ? CopG_antitoxin BrnT_toxin IF-2 GTP_EFTU_D2

JAUSEJ010000531.1:289-4176
Dehalococcoidia bacterium

CopG_antitoxin BrnT_toxin PIN PhdYeFM_antitox ? ?

JAUSEK010000026.1:33162-35584
Candidatus Omnitrophota bacterium

Acetyltransf_7 Sigma70_r2 Epimerase Methyltransf_25 PrmA Methyltransf_11 Glycos_transf_1 Glyco_transf_4 GGDEF GDE_C Glyco_hydro_77 Csm1_N CopG_antitoxin BrnT_toxin ChlI Mg_chelatase Mg_chelatase_C DUF212 YkuD LysM ACCA Radical_SAM RlmN_N Aminotran_5 BPL_C BPL_LplA_LipB HTH_11

JAUSDU010000572.1:5241-7524
Dehalococcoidia bacterium

CobD_Cbib DBI_PRT CbiA GATase_3 Transp_cyt_pur ? BrnT_toxin N6_Mtase HsdM_N

JAUSCW010000424.1:183-3746
Syntrophales bacterium

BrnT_toxin CopG_antitoxin 23S_rRNA_IVP Bac_transf Bac_transf HTH_24 HATPase_c_4 Y1_Tnp HTH_3

JAUXWQ010000034.1:21071-28867
Coxiellaceae bacterium

IMPDH Hydrolase_like Formyl_trans_N Formyl_trans_C DprA_WH DNA_processg_A SAM_SMF DUF494 Zn_ribbon_Top1 Zn_ribbon_Top1 Zn_ribbon_Top1 Topoisom_bac Toprim CopG_antitoxin BrnT_toxin Sua5_yciO_yrdC

JAUYEE010000159.1:5764-14464
bacterium

? PALP ? DUF6946 ? Hexapep Hexapep CopG_antitoxin BrnT_toxin ParE_toxin RelB-like ?

JAUYEC010000061.1:1-13203
Candidatus Omnitrophota bacterium

Rce1-like PEMT CheB_methylest BrnT_toxin CopG_antitoxin ?

JAUYDK010000182.1:1363-3748
Candidatus Omnitrophota bacterium

? ? CopG_antitoxin BrnT_toxin ?

JAUYBU010000214.1:950-13734
Bryobacterales bacterium

MFS_1 MFS_1 QueC Radical_SAM OMP_b-brl_4 DUF2442 DUF4160 AIRC BrnT_toxin CopG_antitoxin AIRC ? ? AIPR

JAUYBR010000532.1:1139-5360
Chloroflexota bacterium

? ? BrnT_toxin MCM ? ? TIR

JAUYAH010000090.1:5361-12706
Thermodesulfovibrionales bacterium

LeuA_dimer HCS_D2 HMGL-like CDP-OH_P_transf PS_Dcarbxylase KARI_C KARI_N adh_short AAA_26 VitK2_biosynth VitK2_biosynth BrnT_toxin CopG_antitoxin

JALYJW010000011.1:36574-51348
Acidobacteriota bacterium

? Lip_A_acyltrans Glyco_trans_2_3 Glycos_transf_2 Glyco_trans_1_2 Glycos_transf_2 PIG-L DUF3369 GGDEF TIM BrnT_toxin CopG_antitoxin SecG ? Biotin_carb_N CPSase_L_D2 Biotin_carb_C Biotin_lipoyl STY-kinase TPRs PTPS

JAUPKS010000103.1:7586-26566
Planctomycetota bacterium

Secretin ? Amidase_2 TPRs TPRs TPRs TPRs TPRs TPRs TPRs Receiver PAS PAS PAS PAS PAS PAS GAF_2 PAS GGDEF EAL PIN AF2212-like BrnT_toxin CopG_antitoxin ? ? ? Cytochrom_C Cytochrom_C Cytochrom_B_C Cytochrome_B Rieske

JALYUL010001083.1:18-2514
Pseudomonadota bacterium

BrnT_toxin DDE_5 MFS_1

JALYZD010000530.1:1-9377
Chloroflexota bacterium

MFS_3 PD40 PD40 Peptidase_S9 Glu_cyclase_2 BrnT_toxin CopG_antitoxin MR_MLE_N MR_MLE_C ? 2-oxoacid_dh E3_binding NUDIX Zn_ribbon_NUD NUDIX-like DUF6789 Oxidored_q2

JAVHLC010000019.1:5031-15996
Thermodesulfobacteriota bacterium

YhcG_C DUF1016_N Abi_C ? ? ? ? CopG_antitoxin BrnT_toxin Resolvase HTH_7 ? HAMP His-Kinase His-Kinase Receiver Trans_reg_C ? ? ?

JARLGA010000129.1:1-6884
Terracidiphilus sp.

FTSW_RODA_SPOVE Mur_ligase_C Mur_ligase_M MurD-like_N Methyltransf_21 Glycos_transf_4 MraY_sig1 ? BrnT_toxin CopG_antitoxin KARI_C KARI_N

JAVVFW010000240.1:1-2551
Armatimonadota bacterium

DUF4058 DZR_2 Pribosyltran GatC ? BrnT_toxin

JARYGU010000002.1:339992-350154
Shinella sp.

? Acetylase ? ? LysR_substrate HTH_1 Fer2 RACo_linker Raco_middle RACo_C_ter CopG_antitoxin BrnT_toxin Pterin_bind MTHFR DUF5981 Virulence_fact Formyl_trans_N DUF1638 ?

JAXMKS010000061.1:1-1582
Candidatus Curtissbacteria bacterium

DUF4389 BrnT_toxin CopG_antitoxin ?

JAXMJV010000931.1:1-6329
Candidatus Binatia bacterium

NMT1_3 DctM Peptidase_M24 Creatinase_N ? BrnT_toxin ? ?

JAPDQB010000010.1:101750-115942
Candidatus Zhuqueibacter tengchongensis

COX1 COX2_TM COX2 Cytochrome_CBB3 SCO1-SenC ? ? Cytochrome_CBB3 CopG_antitoxin BrnT_toxin ActD NrfD Fer4_7 Fer4_7 RES Xre-like-HTH Xre_MbcA_ParS_C Cytochrom_CIII Cytochrome_C7 S-methyl_trans

JAYEFZ010000167.1:255-3908
Candidatus Auribacterota bacterium

BrnT_toxin CopG_antitoxin ? ? Methyltransf_11

JAXDID010000118.1:1957-5831
Thermoanaerobaculia bacterium

DinB ? SAP ? Ble-like_N ? HTH_31 BrnT_toxin CopG_antitoxin ?

JAYELH010000075.1:114-5261
Chloroflexota bacterium

MM_CoA_mutase BrnT_toxin ? PFK tRNA_edit N-OB HHH_6

SHVQ01000028.1:7546-23288
Rhodospirillales bacterium

Formyl_trans_N Arginase tRNA-synt_1e DALR_2 CysS_C tRNA-synt_1c Anticodon_2 CN_hydrolase NAD_synthase ? CopG_antitoxin BrnT_toxin ? PEMT QRPTase_N DAHP_synth_2 Arginase Creatinase_N Peptidase_M24 Pyr_redox_2 Pyr_redox_dim

JAAAOH010000206.1:1-8582
Spirochaetes bacterium

DUF6051 TetR_N MMPL MMPL LolA_like ? Abhydrolase_6 CopG_antitoxin BrnT_toxin

JAACWQ010000002.1:137473-11474
Candidatus Paceibacterota bacterium

BrnT_toxin ? CopG_antitoxin BrnT_toxin Acetyltransf_10 Acetylase ? BrnT_toxin CopG_antitoxin ? Methyltransf_11 RRM_1 ATPgrasp_N ATP-grasp_4 LAL_C2 Dala_Dala_lig_C CitMHS CBS CBS

JAACUO010000443.1:22738-28206
Armatimonadota bacterium

ANKs ANKs ANKs ANKs ANKs ANKs ANKs ? ? ? PhdYeFM_antitox CopG_antitoxin BrnT_toxin PhdYeFM_antitox PemK_toxin ?

JAACTU010000067.1:1-4825
Nitrospirae bacterium

ATPase_gene1 ATP-synt_I PGI AdoMet_Synthase CopG_antitoxin BrnT_toxin MFS_1 MFS_1

WLWO01000003.1:112569-127585
Candidatus Latescibacterota bacterium

FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 Sulfatase SGSH_C Epimerase AP_endonuc_2 Aminotran_3 2-Hacid_dh_C ? BrnT_toxin CopG_antitoxin APH_6_hur DHDPS Methyltransf_25 GFO_IDH_MocA GFO_IDH_MocA_C3 MatE MatE NTP_transf_9

JAAYXB010000011.1:59700-76442
Bacillota bacterium

? LptD_N OstA_2 LptD LptD_N LptF_LptG Toluene_X YjbH BrnT_toxin CopG_antitoxin Hexapep Hexapep Hexapep Hexapep LpxD OmpH OmpH Omp85 POTRA POTRA POTRA

JAAYCQ010000196.1:2935-2319
Syntrophomonadaceae bacterium

ParE_toxin PhdYeFM_antitox ? CopG_antitoxin BrnT_toxin

JAAZNA010000527.1:1-3137
Spirochaetes bacterium

Zip Zip BrnT_toxin CopG_antitoxin ? Ribosomal_S2 EF-Ts_N EF_TS EF_TS

JAADII010000062.1:15846-28024
Chloroflexota bacterium

ParE_like DUF2281 DUF3048_C DUF3048 CaMKII_AD SH3_3 PA14 PA14 PKD_4 PKD_4 META DUF1949 UPF0029 GATase BrnT_toxin CopG_antitoxin DNA_processg_A ZnR HhH-GPD NUDIX_4 RelE ? QTMAN_N Glycos_transf_1

JABLXP010000028.1:12018-10652
Bacillota bacterium

? ? CopG_antitoxin BrnT_toxin DUF5128 Claudin_3 C19orf12 Cu_amine_oxidN1 Cu_amine_oxidN1 UvrD-helicase UvrD_C_2 ? ?

JABMRD010000032.1:7321-16626
Actinomycetota bacterium

? Band-7 Phasin Phasin ? Methyltransf_31 HEATs HEATs CopG_antitoxin BrnT_toxin HTH_3 Gp49 TadE ? ?

MEKX01000051.1:18948-31424

Acidobacteria bacterium RIFCSPLOWO2_12_FULL_54_10

PilZ Sigma54_AID Sigma54_CBD Sigma54_DBD ABC_tran BCA_ABC_TP_C LptC LptD_N LptD_N LptD_N TPR_NPHP3 TPR_NPHP3 ACAS_N AMP-binding AMP-binding_C CopG_antitoxin BrnT_toxin YoeB_toxin ORF6N T5orf172 PhdYeFM_antitox

MFGX01000122.1:24971-16814

Candidatus Fraserbacteria bacterium RBG_16_55_9

DciA Proteasome MoxR-AAA AAA_2 ClpB_D2-small BrnT_toxin CopG_antitoxin tRNA_SAD tRNA-synt_2b HGTP_anticodon Acetylase Ribosomal_S30AE Ribosom_S30AE_C ThiI NFACT-R_2 UvrD-helicase UvrD_C RNA_pol_Rpb1_1 RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 Glyco_transf_4 Glycos_transf_1

MFNZ01000027.1:12584-26657

Candidatus Levybacteria bacterium RIFCSPHIGHO2_02_FULL_37_13

Transgly Transpeptidase Ribul_P_3_epim LytR_C PT SpoIID bPH_2 DUF5657 LytR_C BrnT_toxin CopG_antitoxin ? SmpB Fic HNH_repeat GIY-YIG ?

MGGO01000020.1:25802-36464

Candidatus Woesebacteria bacterium RIFCSPHIGHO2_01_FULL_44_10

PMT_2 ? Polysacc_synt_3 ? ? PMT_2 CopG_antitoxin BrnT_toxin Glyoxalase ? Glycos_transf_2 Glyco_trans_2_3 Glycos_transf_2

MGQS01000040.1:3530-3137
Deltaproteobacteria bacterium RBG_16_54_11

zf-HC2 Sigma70_r4_2 Sigma70_r2 HEPN NTP_transf_2 BrnT_toxin CopG_antitoxin

MGUQ01000055.1:1544-8883
Elusimicrobia bacterium RIFCSPLOWO2_01_FULL_54_10

SMC_ScpB CpXC DUF6981 ? DsrE Cys_rich_CPXG CopG_antitoxin BrnT_toxin SpoU_methylase CHAD Pirin Pirin_C ? ? Ubie_methyltran Oxidored_q4

MGVN01000181.1:18386-28190

Elusimicrobia bacterium RIFOXYB2_FULL_49_7

CHASE4 HAMP PAS Hpt His-Kinase GGDEF PAS MFS_1 PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I PDDEXK_1 PIN VapB_antitoxin CopG_antitoxin BrnT_toxin HTH_3 adh_short CopG_antitoxin

MHAI01000004.1:5563-7301

Ignavibacteria bacterium RIFCSPHIGHO2_02_FULL_56_12

? Y1_Tnp ? ERCC4 HHH_5 BrnT_toxin CopG_antitoxin DUF4386 Beta-lactamase

MHHM01000015.1:1291-4813

Omnitrophica WOR_2 bacterium RIFOXYC2_FULL_45_15

Anth_synt_I_N Chorismate_bind DUF4143 ? BrnT_toxin CopG_antitoxin

MNUP01000123.1:1007-7682

Candidatus Desantisbacteria bacterium CG1_02_49_89

CopG_antitoxin BrnT_toxin ? ParE_toxin CoA_binding Put_DNA-bind_N Fer4_22 Fer4_7 DHODB_Fe-S_bind NAD_binding_1 FAD_binding_6 Fer4_11 NADHdh

MNVQ01000016.1:167-6807
Candidatus Omnitrophica bacterium CG1_02_41_171

PMT_2 Glycos_transf_2 TsaD GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C BrnT_toxin CopG_antitoxin Y1_Tnp

MNXB01000066.1:7129-9135

Elusimicrobia bacterium CG1_02_37_114

? BrnT_toxin CopG_antitoxin BPD_transp_1 BPD_transp_1 Glyco_hydro38C2 Glyco_hydro_38C Glyco_hydro_38 Alpha-mann_mid Glyco_hydro_38N Gp49 HTH_3 URO-D

MNXB01000010.1:10676-20784

Elusimicrobia bacterium CG1_02_37_114

? Guanylate_kin Receiver Receiver Histone ? CopG_antitoxin BrnT_toxin DUF16 Wbp11 ? ? DHR-2_Lobe_B ParA-Soj-NTPase Fer4_NifH ? ?

MWAT01000141.1:162-8527
bacterium ADurb.Bin363

CopG_antitoxin BrnT_toxin Glyco_transf_4 Glycos_transf_1 Glycos_transf_2 Wzy_C ? PMT_2 Glyco_transf_4 Glycos_transf_1 ?

MWBZ01000010.1:21166-37272
bacterium ADurb.Bin243

Peptidase_M55 AhpC-TSA Methyltransf_25 Peptidase_S8 Fic Fic_N Glyoxalase CopG_antitoxin BrnT_toxin RmlD_sub_bind FecR ? Alginate_exp NYN IspD GHMP_kinases_C GHMP_kinases_N

PCSJ01000224.1:1-3187

Deltaproteobacteria bacterium CG23_combo_of_CG06-09_8_20_14_all_51_20

? FitA-like_RHH PIN DUF4160 BrnT_toxin CopG_antitoxin Abhydrolase_1

PCWE01000016.1:1132-5120

Nitrospirae bacterium CG11_big_fil_rev_8_21_14_0_20_41_14

? ? ? ? ? CopG_antitoxin BrnT_toxin ParA-Soj-NTPase

PFDK01000024.1:2554-6390

Candidatus Pacebacteria bacterium CG10_big_fil_rev_8_21_14_0_10_56_10

CopG_antitoxin BrnT_toxin PIN PhdYeFM_antitox CopG_antitoxin PIN AbiEii AbiEi_4 ?

PEYP01000066.1:44-6359

bacterium (Candidatus Stahlbacteria) CG08_land_8_20_14_0_20_40_26

? AAA_14 DUF4143 ? CopG_antitoxin AAA_14 DUF4143 AAA_14 DUF4143 ? BrnT_toxin

PEUZ01000053.1:655-15145

Candidatus Roizmanbacteria bacterium CG03_land_8_20_14_0_80_36_21

ELFV_dehydrog ELFV_dehydrog_N Trns_repr_metal ? DUF6377 HTH_Tnp_1 HTH_21 rve rve_3 Dockerin_1 BrnT_toxin CopG_antitoxin ATP-synt_ab Rho_RNA_bind ABC2_membrane_6 ABC2_membrane_6 ABC_tran ?

PETQ01000144.1:156-13786
Candidatus Desantisbacteria bacterium CG02_land_8_20_14_3_00_49_13

DUF6785 DUF6784 DUF6290 ParE_toxin PIN ? BrnT_toxin CopG_antitoxin PUA_4 Methyltrans_RNA PfkB DcpS_C PNPase tRNA-synt_1 Anticodon_1 zf-FPG_IleRS zf-dskA_traR

PFTU01000656.1:1-8399

Armatimonadetes bacterium CG_4_9_14_3_um_filter_66_14

T2SSF Sulfatase Sulfatase CopG_antitoxin BrnT_toxin ADH_zinc_N ADH_N HTH_17 PemK_toxin ? SPASM Radical_SAM TPRs Spermine_synth

PFEL01000123.1:4170-4000

Candidatus Shapirobacteria bacterium CG10_big_fil_rev_8_21_14_0_10_38_14

Sigma70_r1_1 Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind CopG_antitoxin BrnT_toxin ? MASE2 ?

MUCR01000065.1:31558-38288
SAR202 cluster bacterium Io17-Chloro-G4

Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10 Acyl-CoA_dh_2 Acyl-CoA_dh_M Acyl-CoA_dh_N Acyl-CoA_dh_2 Acyl-CoA_dh_N ? DinB_2 ? CopG_antitoxin BrnT_toxin HMGL-like

PXPI01000060.1:3445-7061

Cyanobacteria bacterium SW_9_44_58

? Cons_hypoth95 BrnT_toxin CopG_antitoxin DUF1636 Peripla_BP_2 TonB_dep_Rec_b-barrel ? DUF2064

QEUW01000020.1:42924-57920
Chloroflexi bacterium

PIN TaqI_C Eco57I HTH_5 AHSA1 DUF4177 CLP_protease MFS_1 MFS_1 BrnT_toxin CopG_antitoxin tRNA_SAD tRNA-synt_2c Aldo_ket_red Pps Creatininase ? BPD_transp_1_N BPD_transp_1

PYIZ01000031.1:9666-21168
Candidatus Poribacteria bacterium

AhpC-TSA PUA AA_kinase GHMP_kinases_C GHMP_kinases_N Aminotran_1_2 PSII_BNR PSII_BNR PSII_BNR ? CopG_antitoxin ThiF ? PIN DUF4258 RelB_like_antitoxin MazG DUF1967 MMR_HSR1_Xtn MMR_HSR1 GTP1_OBG BrnT_toxin

QMMD01000805.1:1-3291
Deltaproteobacteria bacterium

? CopG_antitoxin ? AbiEii ? AbiEi_4 AbiEi_1 BrnT_toxin

QMNR01000299.1:2546-4613
Chloroflexota bacterium

BrnT_toxin CopG_antitoxin PseudoU_synth_2 S4 Zn_dep_PLPC Epimerase Cytidylate_kin

QMOG01000258.1:1-3343
Chloroflexi bacterium

? CopG_antitoxin BrnT_toxin MgsA_C AAA_assoc_2 AAA Abhydrolase_11

QNAB01000099.1:1-3171
Candidatus Aenigmatarchaeota archaeon

BrnT_toxin CopG_antitoxin ? Hist_deacetyl MGS

RFIW01000724.1:135-6914
Chloroflexi bacterium

ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C BrnT_toxin ATP-synt_DE_N ATP-synt_DE MreB_Mbl CoA_binding Put_DNA-bind_N Big Pyr_redox_2

RPQU01000480.1:33-2813
Zetaproteobacteria bacterium

GIY-YIG BrnT_toxin N-OB HepT-like CopG_antitoxin

SEEH01001015.1:98-1119
Rhizobiaceae bacterium

BrnT_toxin ? Peptidase_S49

SCTZ01000006.1:10349-6536
Chloroflexota bacterium

Orn_Arg_deC_N BrnT_toxin CopG_antitoxin Dus YgjP-like Trns_repr_metal Lipoprotein_21 YoeB_toxin PhdYeFM_antitox

SMVQ01000078.1:1785-19238
Planctomycetota bacterium

Sigma70_ECF Lcl_C fn3 PQQ_2 GntR ABC_tran ? CopG_antitoxin BrnT_toxin PhoX ? Peptidase_M6 P_proprotein P_proprotein DUF2959 ? Amidohydro_3 HUTI_composite_bact HMGL-like

VBLE01000437.1:1-3015
Deltaproteobacteria bacterium

TPRs YjzC TPR_MalT ? BrnT_toxin ABC_membrane ABC_tran

CP070770.1:1170757-1187454
Deltaproteobacteria bacterium

PgpA Diphtheria_T MMPL MMPL ? Beta-lactamase Fer4_10 Nitroreductase CopG_antitoxin BrnT_toxin ? Fer4_10 SnoaL_4 Amino_oxidase ACR_tran HlyD_D23 RF-1 Peptidase_M28

CP129483.1:1258204-1275062
Anaerolineales bacterium

DUF3054 Ribosomal_L12 N-OB tRNA-synt_2 GreA_GreB_N GreA_GreB PIG-L PilZ CopG_antitoxin BrnT_toxin ? HEATs Caspase Y1_Tnp Alpha-amylase His-Kinase His-Kinase GAF_2 Abhydrolase_1 3keto-disac_hyd ?

NZ_CP097576.1:316322-332206
Microcystis aeruginosa str. Chao 1910

DUF760 Proton_antipo_M Oxidored_q5_N Hexapep Hexapep XhlA Baculo_PEP_C XhlA ABC_tran DUF5677 BrnT_toxin Inositol_P SMC_ScpB NAD_binding_2 NAD_binding_11 Methyltransf_11 DUF6671 S1 S1 S1

NZ_AOHC02000007.1:57610-65619
Leptospira weilii serovar Ranarum str. ICFT

DUF6364 MazE_antitoxin Peptidase_M23 Discoidin RHH_1 PIN CopG_antitoxin PIN LMWPc PIN HTH_3 HigB-like_toxin BrnT_toxin ParE_toxin VapB_antitoxin Fic

NZ_JANFDD010000005.1:212267-223773
Streptomyces parvus

? SpdB GntR UTRA DCD DUF2637 DnaJ_CXXCXGXG FtsK_SpoIIIE FtsK_SpoIIIE Acetylase BrnT_toxin CopG_antitoxin ? ? HNH Pribosyltran Phage_integrase Phage_int_SAM_3

NC_010424.1:1595521-1607563
Candidatus Desulforudis audaxviator MP104C

EVE Cu_amine_oxidN1 CopG_antitoxin BrnT_toxin ? Ldh_1_N Ldh_1_C ? ? ? ? MFS_1 ? ? ? DUF2442

NZ_JALPTI010000001.1:2718892-2734877
Nocardiopsis dassonvillei

Ribosomal_S15 DUF503 YlxR RBFA Dyp_perox_N Dyp_perox_C ? Sigma70_r2 Sigma70_r4_2 Peptidase_M16_C Peptidase_M16 S1 KH_1 RNase_PH_C RNase_PH PNPase RNase_PH_C RNase_PH Flavokinase FAD_syn BrnT_toxin CopG_antitoxin Abhydrolase_6 TruB_C TruB_C_2 TruB_N KH_NusA_C KH_5 S1 NusA_N NusA_N

NC_015573.1:1933591-1946865
Desulfofundulus kuznetsovii DSM 6115

NusB DUF2273 Asp23 Biotin_carb_C CPSase_L_D2 Biotin_carb_N Biotin_lipoyl PYC_OADA HMGL-like HMGL-like HTH_29 CopG_antitoxin FlpD Fer4_6 Fer4_6 Fer4_2 NAD_binding_8 Fer4_9 Succ_DH_flav_C FAD_binding_2 Na_sulph_symp Asp23 SLH Fer4_9 BrnT_toxin

NC_017791.1:153693-170698
Deinococcus gobiensis I-0

DDE_Tnp_1 DDE_5 CRISPR_Cas6 ? ? Beta-Casp ? BrnT_toxin CopG_antitoxin VWA_2 TROVE TROVE ? DDE_5 DUF1349 Abhydrolase_1 WYL WCX DUF4096

NC_019780.1:722803-733167
Dactylococcopsis salina PCC 8305

? SKI Aldo_ket_red SMC_N Glyco_transf_5 Glycos_transf_1 CopG_antitoxin BrnT_toxin MbhE MnhB MbhD PhaG_MnhG_YufB ? MNHE Proton_antipo_M Oxidored_q2 Xre-like-HTH

NZ_BDJL01000019.1:17293-35572
Carboxydothermus islandicus

CopG_antitoxin HKD ATPase_2 Rce1-like Glyco_hydro_76 Thioredox_DsbH PhdYeFM_antitox BrnT_toxin JetA DUF4194 SbcC_Walker_B AAA_29 ? LapA_dom ? JetD_C ?

NZ_JMQM01000001.1:79074-94166
Nitratireductor basaltis

CobW_C cobW Beta-prop_WDR3_1st WD40_Prp19 DUF4147 MOFRL HUTI_composite_bact Amidohydro_1 LysR_substrate HTH_1 CopG_antitoxin XdhC_CoxI XdhC_C Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 Fer2 Fer2_2 FAD_binding_5 CO_deh_flav_C Transthyretin Cupin_2 Cupin_3 Ureidogly_lyase OHCU_decarbox Polysacc_deac_1 Creatininase BrnT_toxin

NZ_CDRZ01000239.1:33810-12888
Syntrophaceticus schinkii

? DUF6922 CopG_antitoxin RelB Spectrin Helicase_C DEAD ABC_tran Helicase_C AEP-TOTE BrnT_toxin WYL WCX HTH_21 ? ? ?

NZ_FPCB01000007.1:128574-143211
Methylobacterium sp. 174MFSha1.1

CSD HTH_32 DDE_3 DNA_pol_A_exo1 HRDC tRNA-synt_2 N-OB Glyco_trans_2_3 Glycos_transf_2 Acyl_transf_3 CopG_antitoxin BrnT_toxin BRCT HHH_2 DNA_ligase_ZBD DNA_ligase_OB DNA_ligase_aden Nlig-Ia ? ? WYL MerR_1 MerR-DNA-bind RNase_T MFS_1

NZ_AP014815.1:2542184-2555998
Geminocystis sp. NIES-3708

TniQ Methylase_S Methylase_S ? BrnT_toxin Ribosomal_S14 MerR_1 ? ? N6_Mtase N6_Mtase DUF2203 HsdM_N EcoEI_R_C HSDR_N_2 ResIII Helicase_C HTH_24 SLFN_AlbA_2 HATPase_c_4 MoaC

NZ_FOYM01000059.1:376-5603
Desulfoscipio geothermicus DSM 3669

Fer4_7 ? CopG_antitoxin BrnT_toxin PIN Toprim ? cNMPBD T1RH-like_C Toprim_Crpt

NZ_LN890655.1:2057975-2075155
Candidatus Promineifilum breve

? HD CBM96 GH113 CopG_antitoxin BrnT_toxin HGTP_anticodon tRNA-synt_2b tRNA_SAD ? FMN_dh Transpeptidase PBP_dimer MecA_N ? zf-HC2 Bactofilin ? Sigma70_r4_2 Sigma70_r2 HGTP_anticodon tRNA-synt_2b tRNA_SAD

NZ_PPPD01000001.1:1584351-1598768
Deinococcus koreensis

Aminotran_1_2 EthD GGDEF Rce1-like HTH_20 BrnT_toxin CopG_antitoxin TPRs CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 PIN AAA_23 PHP MFS_1 ? ? ?

NZ_CP136419.1:2218891-2227205
Neomoorella humiferrea

Na_H_Exchanger UPF0236 ? ? ? ? BrnT_toxin CopG_antitoxin ? ? ? ? DUF2442 DUF5615 DUF433

NZ_QPMH01000011.1:42551-54744
Ferruginivarius sediminum

Ribosomal_L20 Ribosomal_L35p MarR_2 Glycos_transf_1 Glyco_transf_4 Hydrolase_4 BrnT_toxin CopG_antitoxin Epimerase ? GST_C GST_N_3 Zip HGTP_anticodon tRNA-synt_2b tRNA_SAD TGS ? TPRs TPRs TPRs IF3_C IF3_N

NZ_AP018449.1:1248520-1275275
Methylomusa anaerophila

? ? TPS Fil_haemagg Fil_haemagg Fil_haemagg_2 Fil_haemagg_2 Fil_haemagg_2 ? CopG_antitoxin ADDB_N UvrD_C PDDEXK_1 ? ? EVE His-Kinase His-Kinase PAS Calcineurin ? BrnT_toxin UvrD-helicase UvrD_C UvrD_C PDDEXK_1 ?

NZ_ML142861.1:41448-56665
Phytoactinopolyspora endophytica

tRNA-synt_1b PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I GalP_UDP_tr_C GalP_UDP_transf Epimerase GHMP_kinases_C GHMP_kinases_N GalKase_gal_bdg CopG_antitoxin BrnT_toxin NUDIX ZnR LacI Peripla_BP_3 HAD_2 LacI Sulfatase SGSH_C adh_short TetR_C_16 TetR_N DeoRC HTH_DeoR

NZ_QFFZ01000001.1:71589-85074
Pelotomaculum propionicicum

HTH_24 AsnC_trans_reg SSF PG_binding_1 Hydrolase_2 DsrC SPOUT_MTase CopG_antitoxin Zn_Tnp_IS91 EPSP_synthase Peptidase_MA_2 TPRs LysM G5 Peptidase_M23 Radical_SAM SPASM SCIFF YPEB_N YPEB_PepSY1-2 PepSY BrnT_toxin

NZ_VWNA01000001.1:2584610-2601520
Segnochrobactrum spirostomi

EamA EamA ? Adenylsucc_synt MS_channel_1st MS_channel_2nd MS_channel_3rd ? ACT PGDH_inter 2-Hacid_dh_C ? BrnT_toxin CopG_antitoxin OMP_b-brl PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I Nitroreductase FCD GntR GAF_2 GGDEF Rho_N DUF6496

NZ_JACIGK010000082.1:1-1503
Roseospira visakhapatnamensis

? CopG_antitoxin ArdcN BrnT_toxin

NZ_AP025032.1:894404-908378
Aureimonas sp. SA4125

ADH_zinc_N ADH_N GTP_cyclohydro2 DHBP_synthase QueF TerC HTH_1 LysR_substrate Complex1_51K NADH_4Fe-4S BrnT_toxin CopG_antitoxin Fer2_4 NADH-G_4Fe-4S_3 Fer4_7 Molybdop_Fe4S4 Molybdopterin Molydop_binding FdhD-NarQ FdsD NodS PIG-L Acyl-CoA_dh_1 2Fe-2S_thioredx

NZ_CP082909.1:4155640-4171028
Endozoicomonas sp. 4G

? Aminotran_5 ? DUF1388 DUF2288 RF-1 Tim44 BrnT_toxin CopG_antitoxin Cas12f1-like_TNB OrfB_IS605 HTH_OrfB_IS605 Tnp_DNA_bind DDE_Tnp_1 ? ? Cas12f1-like_TNB OrfB_IS605 Resolvase

NZ_AP026562.1:508236-524502
Deinococcus aetherius

Abhydrolase_1 ? ADH_N ADH_zinc_N_2 ? ? CopG_antitoxin BrnT_toxin RNase_T Big CBM_4_9 CelD_N Glyco_hydro_9 Collagen_bind_2 Apiosidase CBM60 Lipase_GDSL_2 HMA HMA E1-E2_ATPase Hydrolase HMA ? Trns_repr_metal DUF1064

NZ_JASNGB010000002.1:38624-48141
Deinococcus rhizophilus

adh_short ? NUDIX DnaB DnaB_C PilA4 PilA4 N_methyl CopG_antitoxin BrnT_toxin DUF1802 DUF5057 Methyltransf_25 NAD_Gly3P_dh_C NAD_Gly3P_dh_N ? ? ATP-synt_D DUF1802

NZ_JAGGSG010000131.1:1646-12757
Thermococcus sp.

DUF3450 HTH_24 Phage_integrase HTH_17 HTH_17 ? DnaB DnaB_C ? RHH_1 PC4 HTH_Tnp_1_2 BrnT_toxin CopG_antitoxin Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_C ? ?

NZ_JAVVGM010000153.1:1-4999
Thermus sp.

Peptidase_S24 N-OB HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP ? IMS BrnT_toxin

NZ_JAXCWK010000012.1:6268-24984
Sphingomonas sp. CFBP9019

HsdM_N N6_Mtase DUF4145 REase ResIII Helicase_C EcoEI_R_C HTH_8 AAA_lid_14 NtrC-AAA GmrSD_C GmrSD_N DUF6575 ? CopG_antitoxin BrnT_toxin ? DUF547 TrwB_AAD_bind MobA_MobL ? ? RPA

NZ_CP109000.1:5524424-5537841
Streptomyces sp. NBC_00704

DDE_Tnp_4 HTH_Tnp_4 ? RES HEPN_RES_NTD1 HTH_3 ? ? SLATT BrnT_toxin CopG_antitoxin ? HTH_31 DUF5753 His-Kinase Sigma70_r4 PP2C_2 ?